

CodeFest Datathon 2026 - Task Brief

1) Overview

Climate change is transforming global temperatures, energy systems, emissions, and economic policy at an unprecedented pace. As governments and businesses accelerate the transition toward low-carbon economies, the ability to analyze complex climate and energy data has become essential for informed decision-making.

In CodeFest Datathon 2026, your team will explore a multi-source climate dataset spanning 2000 - 2026 and develop data-driven solutions that uncover meaningful trends, build predictive models, and propose innovative products that support the global energy transition.

Rather than solving a single prediction problem, this challenge emphasizes exploratory analytics, machine learning, feature engineering, forecasting, and business innovation.

2) Background Story

Imagine you are part of a climate analytics consultancy advising governments, energy companies, financial institutions, and sustainability leaders.

Your clients need answers to questions such as:

- How are carbon markets reacting to climate policies and extreme weather?
- Which countries are transitioning fastest toward renewable energy?
- How will different energy strategies affect future CO₂ emissions?
- Can climate data be transformed into commercially valuable products?

Using historical climate, emissions, energy mix, and carbon pricing data, your team will generate actionable insights and predictive models that help stakeholders make evidence-based decisions.

3) What You Will Be Given

Participants will receive **five CSV datasets** containing historical climate and energy information.

File	Description
carbon_prices_daily.csv	Daily carbon prices across five major emissions trading systems
climate_events.csv	Major climate, policy, and extreme weather events
co2_emissions_yearly.csv	Annual CO ₂ emissions and emissions intensity for 50 countries
energy_mix_yearly.csv	Country-level energy generation mix (coal, oil, gas, renewables, etc.)
temperature_anomaly_monthly.csv	Monthly temperature anomalies and atmospheric CO ₂ concentrations

All datasets are provided for use during the competition. **No external datasets or pre-trained models are permitted.**

4) Challenge Objectives

Question 1 - Predictive Modeling for Climate and Energy Data

1.1 Forecasting Carbon Price Prediction

Build a model to forecast daily market carbon prices for the next 30 trading days. You must justify your modeling approach (classical statistical models vs. ML algorithms) and evaluate performance on a test set using RMSE or MAPE.

1.2 Predicting CO₂ Emissions from Energy Mix

Build a regression model predicting `co2_per_capita_t` based on a country's energy mix profile (fuel shares across coal, oil, gas, and renewables). You will be evaluated

on data wrangling completeness, feature engineering, and predictive accuracy (R^2 /RMSE).

Question 2 - Cross-Dataset Feature Engineering - Carbon Price Drivers

Test the hypothesis that carbon prices respond to real-world climate events or policy shifts - a claim often assumed but rarely verified.

Use `carbon_prices_daily.csv` with `climate_events.csv` to engineer event-proximity features (eg: days since last major event, days until next known policy date). Then prove that these features add predictive value, by benchmarking a price-prediction or up/down price-movement classification model with these features against an identical baseline model without them.

Question 3 - Renewable Energy Transition Scenario Modelling

3.1 Energy Mix to CO₂ Relationship

Analyze the data to uncover global energy-transition patterns from 2000 to 2026. Create a visualization that maps energy mix shares (coal, oil, gas, renewables, nuclear) against CO₂ emissions footprints across the countries.

Identify and categorize any distinct transition patterns or cluster archetypes that emerge.

Identify the defined energy-mix trajectories for the past years:

- Business-as-Usual: The energy mix continues following its current trend.
- Moderate Transition: Renewable energy gradually replaces fossil fuels.
- Accelerated Transition: Renewable energy adoption increases significantly.

You must justify your choice of visualization/s used.

3.2 Forecasting - Future Emissions

Project annual CO₂ emissions for each selected country or region across the 2026 - 2030 forecast window. Generate annual CO₂ emissions forecasts (total or per capita) for 2026 through 2030.

Explicitly mention all underlying assumptions, key parameter choices, and assumed annual rates of change driving your model.

3.3 Key Insights & Findings

Highlight the most important discoveries from the data.

Translate technical findings into meaningful business or real-world insights.

4. Pitch your Product

Translate your technical data analysis and models into a viable, data-driven commercial product or service that addresses real-world Environmental, Social, and Governance (ESG), Financial, or regulatory challenges.

Core Deliverables:

- **Value Proposition & Target Market:** Identify a specific target customer (e.g: ESG fund managers, corporate sustainability officers, commodities traders, municipal planners) and the pain point your solution solves.
- **Solution Architecture Diagram for the Proposed Product**
- **Data & Predictive Modelling:** Explain how the dataset (emissions, energy mix, climate events, carbon prices) and your predictive models power the core functionality of your solution.
- **MVP & Technical Proof-of-Concept:** Demonstrate a working, low-fidelity prototype or functional dashboard (e.g: a Carbon Shock Alert System, Climate Risk Scoring API, or ESG Transition Screener).
- **Commercial Strategy:** Define your monetization strategy (e.g: SaaS tier, API call billing, enterprise licensing) and pitch the business viability alongside your technical solution.

5) Dataset Schema

Note: Please refer to the Data_Dictionary.xlsx file for the dataset schema.

6) Submission Requirements

Submit a single compressed folder containing:

- **Teamname_notebook.ipynb** - Complete notebook with preprocessing, analysis, modelling, and visualizations
- **Teamname_presentation.pptx** – Final presentation including the business pitch.
- **README.md (Optional)**– Brief explanation of methodology and assumptions.

Important Notes

- All code must be contained within the submitted notebook.
- Results should be reproducible from the provided datasets.
- Clearly document preprocessing and modelling decisions.
- Be prepared to explain your methodology during judging.

7) Rules & Regulations

- All codes must be original work developed during the Datathon.
- Each team's final presentation **must not exceed 10 minutes**. Presentations will be stopped immediately at the 10-minute mark, so teams should manage their time accordingly.
- Only the provided datasets may be used for model training and analysis.
- External datasets, pretrained models, and pretrained weights are prohibited.
- AI tools (ChatGPT, Copilot, Gemini, etc.) may be used for debugging, documentation, and brainstorming, but teams must implement and understand their own solution.

- Teams must be able to explain their preprocessing, feature engineering, model architecture, and evaluation methodology.
- Any form of plagiarism, collusion, data leakage, or rule violation may result in disqualification.
- Judges' decisions are final.

8) Judging Criteria

Criterion	Focus
Data Analysis	Insight generation and exploratory analysis
Technical Quality	Preprocessing, feature engineering, modelling
Predictive Performance	Accuracy and evaluation methodology
Visualization	Clarity and communication of findings
Business Innovation	Practicality, originality, and commercial viability
Presentation	Technical understanding and storytelling

Good luck and help shape the future of climate intelligence through data.